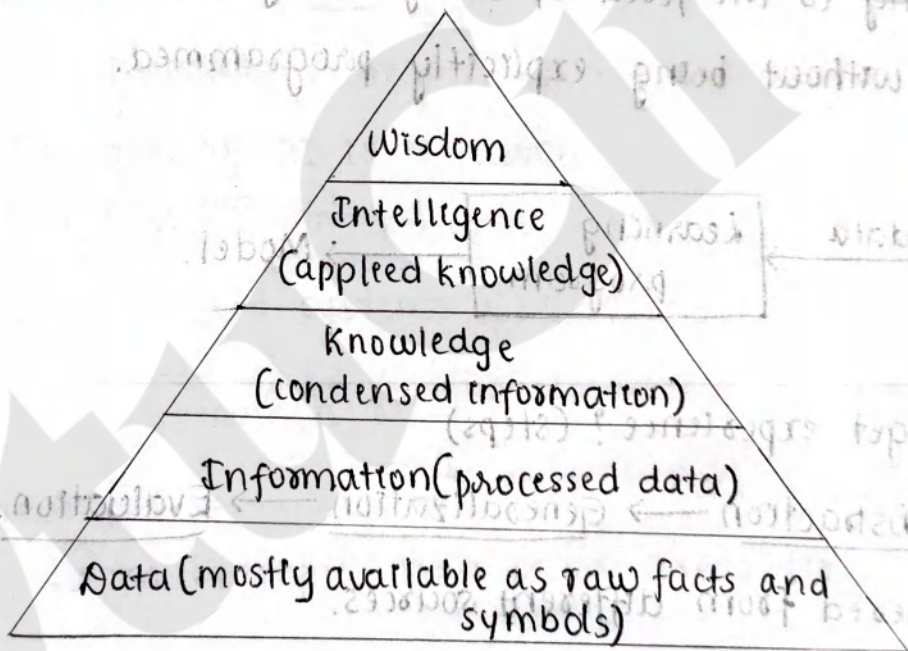


Need for Machine Learning:- (3 Reasons)

- 1) High volume of available data to manage - Companies like Facebook, Twitter and Youtube generate massive amounts of data, and this data keeps doubling every year.
- 2) Cost of storage - storage and hardware costs have dropped, making it easier to collect, process, store and share data.
- 3) Popularity of machine learning - More complex algorithms are now available, especially with deep learning, making machine learning more powerful.

The Knowledge Pyramid:-



Data:- *) All facts are data. Data consists of raw numbers, text or symbols without any context.

Ex:- A store records daily sales numbers like 15, 20, 100

Information:- It is the processed data. It includes patterns, associations (correlations) relationships among data.

Ex:- Sales data shows that product X sells more on weekends.

Knowledge:- Condensed information that provides meaningful insights.

Ex:- product X, sells more in summer.

Intelligence:- Applied knowledge to take action and make decisions.

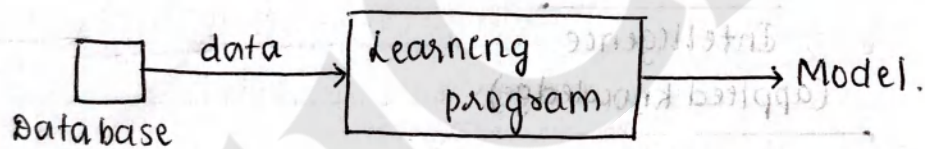
Ex:- The store buys more product X before summer because people buy it more.

Wisdom:- the highest level, representing mature and strategic decision-making.

Ex:- the store creates a new summer product that customers like.

Machine Learning:-

Machine learning is the field of study that gives the computers ability to learn without being explicitly programmed.



How machines get experience? (Steps)

Collection → Abstraction → Generalization → Evaluation.

*> Data is gathered from different sources.

*> Once data is collected, abstract concepts are formed. Abstraction helps create ideas. This is similar to how humans recognize objects. For ex, we know that an elephant looks like without seeing it directly.

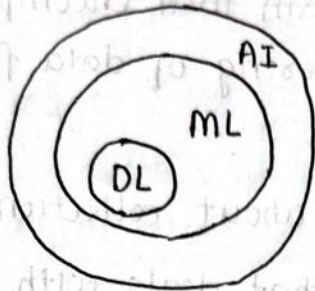
*> Generalization turns abstract ideas into intelligence. It involves ranking concepts, making inferences and forming heuristics (smart guesses). Heuristics are educated guesses used for decision-making.

Ex:- If a person senses danger, they react based on past experiences. Machines follow a similar process.

*> Heuristics usually work, but sometimes they fail. Since heuristics are just "rules of thumb", they need corrections when errors occur. Evaluation checks and improves models to make better decisions.

ML in relation to other fields:-

i) ML and AI:-

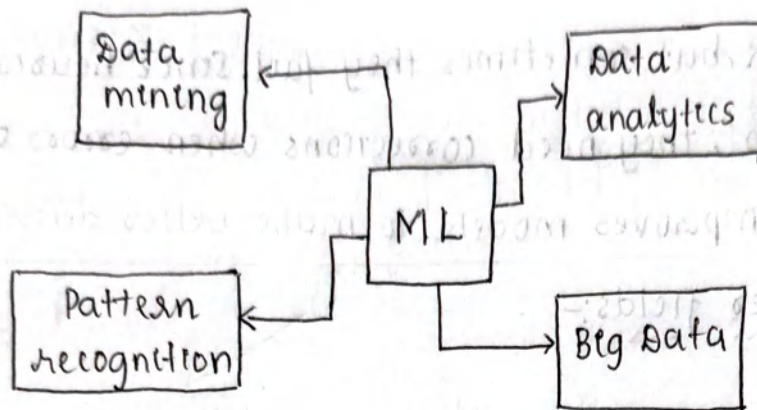


*> ML is an important branch of AI, which is a much broader subject. The aim of AI is to develop intelligent agents. An agent can be robot, humans (or) any autonomous systems. The idea is to develop intelligent systems like human beings.

*> ML is the subbranch of AI, whose aim is to find and extract the patterns for prediction. It is a broad field that includes learning from examples and other areas like reinforcement learning.

*> Deep learning is a subbranch of ML. In DL, the models are constructed using neural network technology. Neural networks are based on the human neuron models.

ii) ML, Data Science, Data Mining and Data Analytics:-



* Data Science is an 'umbrella' term that encompasses many fields. Data science deals with gathering of data for analysis. It is a broad field that includes:

→ Big Data:- Data science concerns about collection of data. Big data is a field of data science that deals with data's following characteristics:-

i) Volume:- Huge amount of data is generated by big companies like facebook, twitter, Youtube.

ii) Variety:- Data is available in variety of forms like images, videos and in different formats.

iii) Velocity:- It refers to the speed at which the data is generated and processed.

→ Big Data is used by many ML algorithms for applications such as language translation and image recognition.

→ * Data Mining:- It aims to extract the hidden patterns that are present in the data, whereas, ML aims to use it for prediction.

→ Data analytics:- Another branch of data science is data analytics.

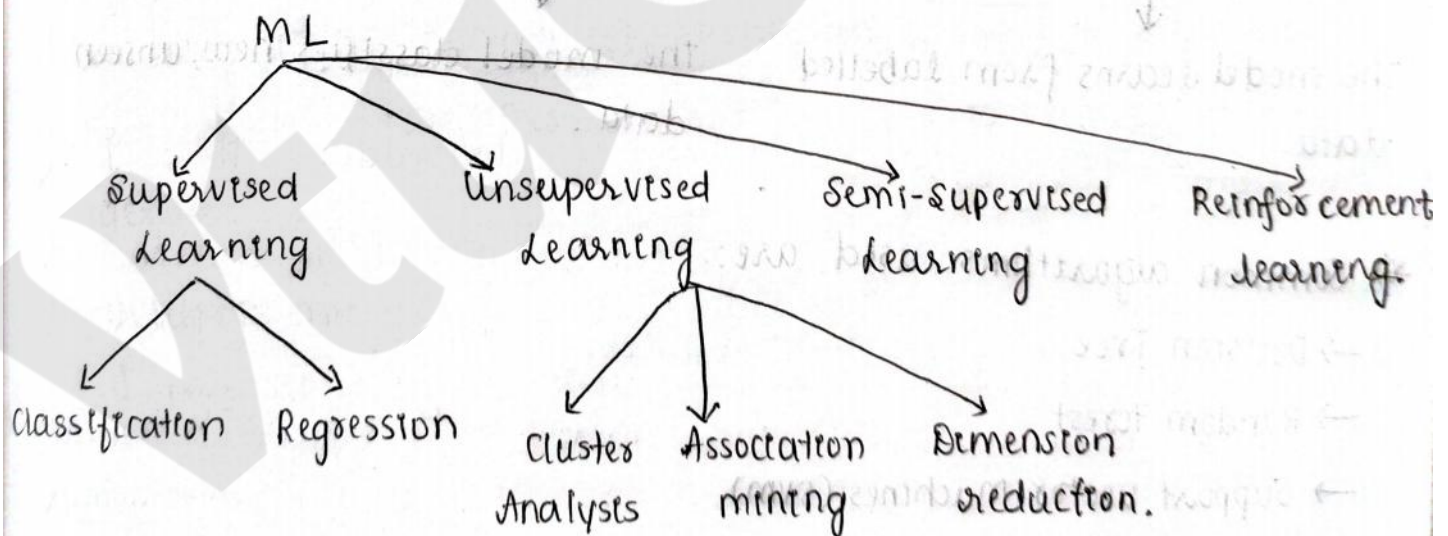
It aims to extract useful knowledge from crude data. There are different types of analytics. Predictive data analytics is used for making predictions.

→ Pattern Recognition:- It is an engineering field. It uses ML to extract the features for pattern analysis and pattern classification.

→ ML and Statistics:-

* Statistics is based on mathematical theories, requiring strong statistical knowledge and hypothesis testing. It looks for patterns and relationships in data through rigorous methods. Whereas, ML makes fewer assumptions and relies on automation to learn from data.

Types of ML:-



Note:- Data can be classified as

1) Labelled :- includes attributes and a target variable (label) that we aim to predict.

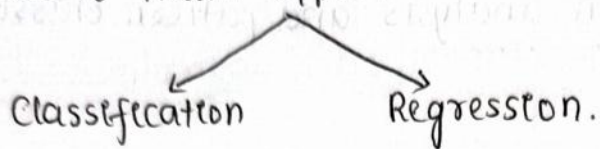
Ex:- Iris-dataset

ii) Unlabelled data:- does not include any attributes.

Supervised learning:-

*> Supervised learning uses labelled data and involves a supervisor who provides this data for model training and testing.

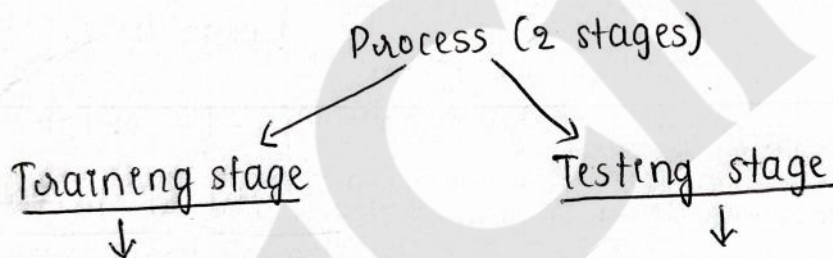
*> It has 2 main types:-



Classification:-

*> Classification is a supervised learning method that predicts labels (discrete values) based on input attributes. The relationship b/w inputs and labels is learned using a classification model.

*>



*> Common algorithms used are:

→ Decision Tree

→ Random Forest

→ Support Vector Machines (SVM)

→ Naive Bayes

→ Artificial neural networks & Deep Learning (eg., CNN, RNN)

Ex:- *> Identifying images of cats and dogs

*> Classifying diseases (or) plant species.

Regression:-

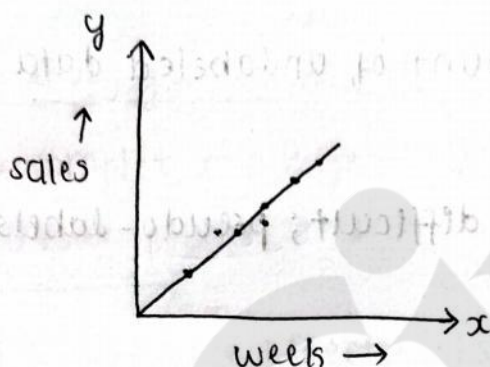
*> Regression predicts continuous numerical values rather than discrete labels.

Ex:- Predicting product sales based on weeks of data using a linear regression model of the form:

$$y = 0.66x + 0.54 \quad (y = mx + c)$$

where $x \rightarrow$ independent variable (week)

$y \rightarrow$ dependent variable (sales)



Unsupervised Learning:-

- *> Learning occurs through self-instruction without a supervisor.
- *> the algorithm identifies patterns and groups similar objects together.

Ex:- Cluster analysis

i) Dimensionality reduction.

Cluster analysis:-

- *> Group similar objects into disjoint clusters based on their attributes.
- *> Used in image segmentation, medical anomaly detection and gene database clustering.

Ex:- A clustering algorithm groups dog and cat images into separate clusters.

Key algorithms:- K-means, PCA, Hierarchical algorithms.

Dimensionality Reduction:-

- *> Reduces the no of features while retaining essential information.
- *> Converts high-dimensional data into lower-dimensional representations.

Semi-supervised Learning:-

- *> uses a mix of a large amount of unlabeled data and some labeled data.
- *> Labelling data is costly and difficult; pseudo-labels are assigned to unlabeled data.
- *> The labeled and pseudo-labeled datasets are combined for training.

Ex:- Autonomous vehicles :- Labeled traffic data helps self-driving cars learn from unlabeled 'real-world' scenarios.

Reinforcement Learning:-

- *> Mimics human learning through interaction with the environment.
- *> An agent takes actions to maximize rewards.
- *> Rewards can be positive (reinforcement) or negative (punishment).

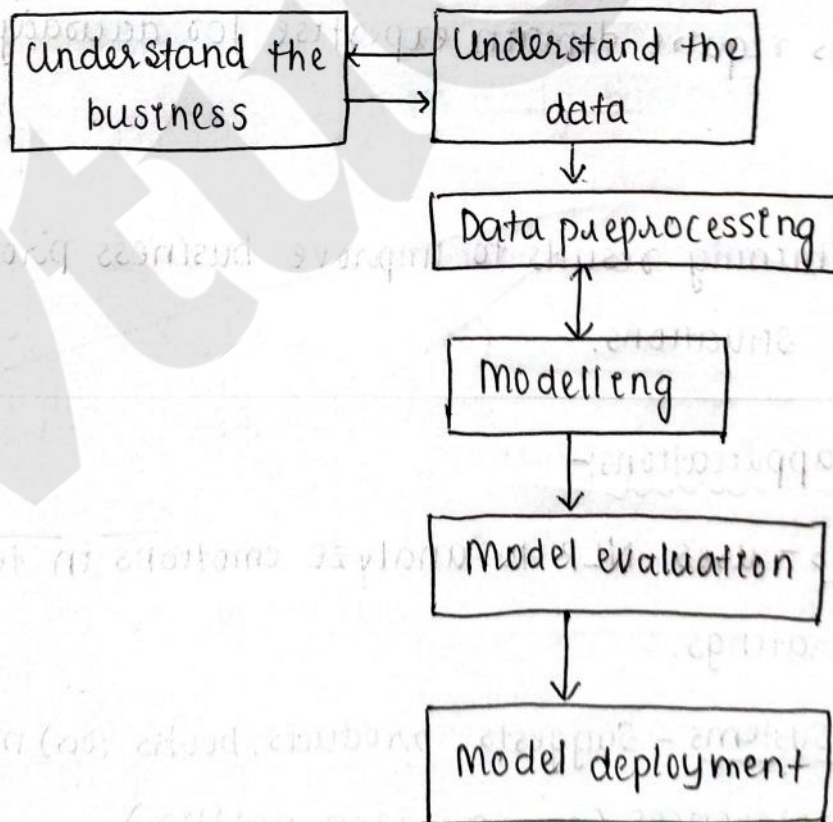
Ex:- A grid-game where the agent learns to navigate from a start position to a goal while avoiding danger.

Note:- It is reward-based + goal-oriented.

Challenges of ML:-

- i) Well-posed vs ill-posed problems:- ML works best when specifications are clear. Some problems lack sufficient data (or) constraints.
- ii) Huge Data requirement:- ML models require large, high-quality datasets without missing (or) incorrect data.
- iii) High Computation Power:- ML, especially with big data, demands GPUs/TPUs for processing. Increased model complexity raises computational needs.
- iv) Complexity of the algorithms:- Selecting, designing and evaluating algorithms is complex and affects performance.
- v) Bias-Variance Tradeoff:- A model should generalize well. Overfitting and underfitting reduce effectiveness.

Machine Learning process:-



Understanding the business:-

- *> Identify objectives and requirements of the business.
- *> Define the problem statement for data mining.

Understanding the data:-

- *> Collect data, analyze characteristics, and formulate hypotheses.
- *> Match patterns to the selected hypothesis.

Preparation of Data:-

- *> Clean and preprocess data to remove missing values.
- *> Ensure data quality for accurate model results.

Modelling:-

- *> Apply data mining algorithms to identify patterns and build models.

Evaluation:-

- *> Assess model performance using statistical analysis and visualization.
- *> Classification tasks require domain expertise for accuracy.

Deployment:-

- *> Implement data mining results to improve business processes (or) address new situations.

Machine Learning applications:-

- Sentiment analysis - uses NLP to analyze emotions in text, such as reviews (or) ratings.
- Recommendation Systems - Suggests products, books (or) movies based on user preferences (e.g., amazon, netflix)

iii) Voice assistants:- AI powered assistants like Alexa, Siri and Google Assistant process voice commands.

iv) Navigation Systems - Google Maps and Uber use ML to optimize routes and reduce travel time.

Data:-

* All facts are data. In computers, data is stored as numbers, text, images, audio (or) video. Some data is directly readable (like text), while other types (like videos) need processing.

* Today, businesses collect and store huge amounts of data, measured in GB, TB, PB (petabytes) and even EB (exabytes). A byte consists of 8 bits, and larger data units are formed by grouping bytes.

* Data is stored in different sources like files, databases, and data warehouses. It can be classified as:

→ Operational Data:- Used in daily business tasks (eg, sales transactions)

→ Non-operational Data:- Used for analysis and decision-making (eg, predicting future sales)

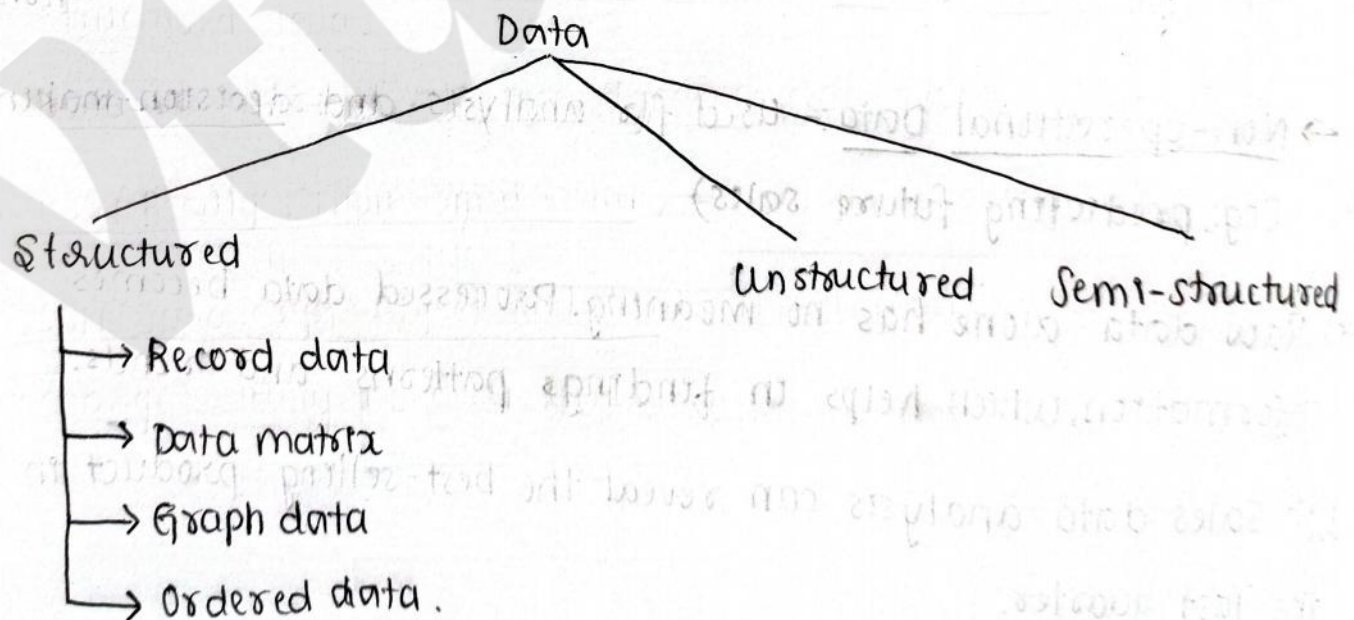
* Raw data alone has no meaning. Processed data becomes information, which helps in finding patterns and trends.

Ex:- Sales data analysis can reveal the best-selling product in the last quarter.

Elements of big data:-

- i) Volume:- Data is generated in massive amounts due to reduced storage costs.
- ii) Velocity:- Data is created and shared at high speed, such as social media updates.
- iii) Variety:- Data comes in multiple forms:-
 - Form:- Text, images, audio, video, maps
 - Function:- sources like conversations, transactions and archive records.
 - Source:- public data, social media and multimodal data.
- iv) Veracity:- Ensuring data is accurate and trustworthy.
- v) Validity:- Checking whether the data is useful for analysts.
- vi) Value:- Extracting meaningful insights from data to help in decision-making.

Types of data:-



Structured data:-

* Structured data is stored in an organized way, like in tables (or) databases. It can be easily retrieved using SQL.

→ Record data:- Data stored in tables with rows (objects) and columns (attributes)

→ Data matrix:- A table with only numeric values, used in mathematical operations.

→ Graph data:- Data showing relationships, like web pages linked to each other.

→ Ordered data:- Data that follows a sequence. Examples:

- Temporal data:- Related to time, like customer purchases during festivals.
- Sequence data:- Ordered data without time stamps, like DNA sequences
- Spatial data:- Data with locations, like maps.

Unstructured data:-

* Unstructured data includes images, videos, audio, documents and blog posts. It makes up about 80% of all data.

Semi-structured data:-

* This data is partly structured and partly unstructured. Examples include XML, JSON, RSS feeds, and hierarchical data.

Data Storage and Representation:-

1) Flat files:-

- * Simple, low-cost way to store data in plain text (ASCII, EBCDIC)
- * Suitable for small datasets.
- * Common formats:-

→ CSV (Comma-separated values):- Data separated by commas, used in spreadsheets.

→ TSV (Tab-separated values):- Data separated by tabs.

2) Database systems:-

*> Stores structured data using a DBMS.

*> Uses SQL for data manipulation

*> Types of databases:-

→ Transactional database:- Stores records of transactions.

→ Time-series database:- Stores data overtime (eg., logs, sales data)

→ Spatial database:- Stores spatial/geographic data in raster format.

3) Other data sources:-

*> World Wide Web (WWW):- A vast online information source for data mining.

*> XML (Extensible Markup Language):- A format for sharing structured data.

*> Data Streams:- Real-time, dynamic data flow.

*> RSS (Really Simple Syndication):- Used for sharing instant updates.

*> JSON (JavaScript Object Notation):- A common format for data exchange, widely used in ML.

Big Data analytics and types:-

The goal of data analytics is to help businesses make decisions by extracting insights from data. Data analysis focuses on analyzing historical data, while data analytics covers the entire process, including data collection and preprocessing.

4-types:-

- 1) Descriptive:- Summarizes and quantifies collected data. It focuses on what happened using statistics.
- 2) Diagnostics:- Explains reasons behind events. It answers "why?" by analyzing causes and effects.
- 3) Predictive:- Forecasts future trends based on data patterns. Uses ML for predictions.
- 4) Prescriptive:- Suggests the best course of action to improve decision-making and minimize risks.

Big Data Analysts Framework:-

Big data analytics framework use a layered architecture. (4 layers)

1) Data Connection Layer:-

- * Handles data ingestion and data connectors.
- * Transforms raw data into structured formats using ETL (Extract, Transform, Load)

2) Data Management Layer:-

- * Preprocesses data and manages storage.
- * Supports parallel execution of queries and data operations.
- * Uses data warehouses and on-demand data retrieval methods.

3) Data Analytics Layer:-

* Uses statistical tests, ML models, and validation techniques.

4) Presentation Layer:-

* Dashboards and applications visualize analytical results.

* Helps interpret ML outputs for decision making.

Big Data Processing Cycle:-

4 steps:-

1) Data collection:-

* Data should be timely, relevant and interpretable.

* Types of data sources:-

→ Open / Public data - Government census, scientific research, healthcare records.

→ Social Media data - Data from platforms like Twitter, Facebook, Youtube

→ Multimodal data - Text, images, videos and audio.

2) Data Preprocessing:-

* Cleans dirty data (incorrect, missing, inconsistent (or) duplicate)

* Removes outliers and corrects formatting issues.

* Ex:- (for bad data)

→ salary = " " → Incomplete data

→ Age = 136 → Outlier (or) typographical error.

3) Applying ML Algorithms:-

*> Uses statistical techniques and ML models to predict and analyze data.

4) Interpretation & Visualization:-

*> Dashboards, graphs and charts help display results for decision-making.

Data preprocessing:-

Real-world data is often dirty and requires preprocessing:-

*> Data cleaning:-

→ Handles missing, duplicate (or) inconsistent values.

→ Removes formatting errors and outliers.

*> Noise reduction:-

→ Filters random fluctuations ⁱⁿ of data.

*> Data wrangling:-

→ Converts raw data into a structured format.

Missing data handling methods:-

- i) Ignore the missing values (if few)
- ii) Fill manually (time-consuming for large datasets)
- iii) Use global constants (eg., "unknown")
- iv) Replace with attribute values (eg., using average)
- v) Use mean imputation (fill missing values with mean)
- vi) Predict missing values (via classification models)

These preprocessing techniques ensure clean, high-quality data for ML.

Removal of Noisy (or) Outlier Data:-

* Noise:- Noise is random variation in data.

* It can be removed using binning, which groups data into bins (buckets) and smooths values using neighbour data.

Binning Techniques:-

1. Smoothing by means $\rightarrow$ replace values with the bin's mean.
2. Smoothing by bin medians $\rightarrow$ replace values with the bin's median.
3. Smoothing by bin boundaries $\rightarrow$ replace values with the nearest bin boundary (max/min)

Q) Consider the following set $S = \{12, 14, 19, 22, 24, 26, 28, 31, 32\}$.

Apply various binning techniques and show the result.

(take depth = 3). { depth $\rightarrow$ no of elements in a bin }

Sol:- Divide the bins $\rightarrow$ (depth 3)

B1 : 12, 14, 19

B2 : 22, 24, 26

B3 : 28, 31, 32

[need to sort the elements
before dividing]

(here already sorted)

Smoothing by means:-

$$B1: \frac{12+14+19}{3} = 15$$

$$B2: \frac{22+24+26}{3} = 24$$

$$B3: \frac{28+31+32}{3} = 30.3$$

$$\therefore B1: 15, 15, 15$$

$$\text{Result: } \{15, 15, 15, 24, 24, 24, 30.3, 30.3,$$

$$B2: 24, 24, 24$$

$$30.3\}$$

$$B3: 30.3, 30.3, 30.3$$

$$=$$

Smoothing by medians:-

$$\text{median}(B1): \text{med}(12, 14, 19) = 14$$

$$\text{med}(B2): \text{med}(22, 24, 26) = 24$$

$$\text{med}(B3): \text{med}(28, 31, 32) = 31$$

$$\therefore B1: 14, 14, 14$$

$$B2: 24, 24, 24$$

$$B3: 31, 31, 31$$

$$\text{Result: } \{14, 14, 14, 24, 24, 24, 31, 31, 31\}$$

$$=$$

Smoothing by boundaries:-

S1:- Sort the no's.

S2:- Fix the min and max boundaries

S3:- Find closest value & replace it for rest of the values, that is,

$$| \text{min} - \text{max} | \text{ (or) } | \text{max} - \text{min} |$$

$$\Rightarrow B1: 12, 14, 19 \Rightarrow B1: 12, 12, 19$$

$$B2: 22, 24, 26 \Rightarrow B2: 22, 22, 26 \quad \left\{ \begin{array}{l} \text{if the diff is same,} \\ \text{consider lower value} \end{array} \right.$$

$$B3: 28, 31, 32 \Rightarrow B3: 28, 31, 32$$

$$\text{Result: } \{12, 12, 19, 22, 22, 26, 28, 31, 32\}$$

Data Integration and transformation:-

Data integration merges data from multiple sources, but it may cause redundancy. The goal is to remove redundancy and improve data processing. Normalization is used to scale data for better algorithm performance.

Normalization methods:-

i) Min-max normalization

ii) Z-score normalization.

Min-max normalization:-

* Min-max normalization scales values to a fixed range.

Formula:-

$$\text{minmax} = \frac{V - \text{min}}{\text{max} - \text{min}} \times (\text{newmax} - \text{newmin}) + \text{newmin}$$

where min, max $\rightarrow$ original data's min and max
newmin, newmax $\rightarrow$ target range.

Note:- Commonly used in neural networks.

Q) Given, $V = \{88, 90, 92, 94\}$. Apply min-max procedure & map the marks to a new range 0-1.

Sol:- slr Sort the given (already sorted)

The min of $V = 88$ and $\text{max} = 94$ and given,
 $\text{newmin} = 0$ and $\text{newmax} = 1$.

consider 88,

$$\Rightarrow \text{minmax} = \frac{88-88}{94-88} \times (1-0) + 0 = 0.$$

consider 90,

$$\Rightarrow \text{minmax} = \frac{90-88}{94-88} \times (1-0) + 0 = 0.33$$

consider 92,

$$\Rightarrow \text{minmax} = \frac{92-88}{94-88} \times (1-0) + 0 = 0.66$$

consider 94,

$$\Rightarrow \text{minmax} = \frac{94-88}{94-88} \times (1-0) + 0 = 1$$

final normalized values :- $\{0, 0.33, 0.66, 1.0\}$

Z-Score normalization:-

this method adjusts values by subtracting the mean and dividing by the standard deviation.

$$V^* = \frac{V - \mu}{\sigma}$$

where μ = mean

σ = standard deviation.

Q) Given, $V = \{10, 20, 30\}$, convert the marks to z-score.

Sol:-

Note:-

$$\sigma = \sqrt{\frac{\sum (x_i - \bar{x})^2}{N-1}}$$

$$\Rightarrow \mu = \frac{10+20+30}{3} = 20.$$

$$\sigma = \sqrt{\frac{(10-20)^2 + (20-20)^2 + (30-20)^2}{3-1}}$$

$$\sigma = \sqrt{\frac{100+100}{2}} = \sqrt{100} = 10$$

$$z\text{-score of } 10 = \frac{10-20}{10} = -1$$

$$z\text{-score of } 20 = \frac{20-20}{10} = 0$$

$$z\text{-score of } 30 = \frac{30-20}{10} = 1$$

$$\therefore z\text{-score of } (10, 20, 30) = (-1, 0, 1)$$

What is the use of z-scores?

→ z-scores are used to detect outliers. If the data value of z-score function is either less than -3 (or) greater than +3, then it is possibly an outlier.

Data reduction:-

Reduces data size while maintaining results. Methods include:

- * Data aggregation
- * Feature selection
- * Dimensionality reduction.

$$\frac{(\bar{x} - x)^2}{1 - n}$$

Descriptive statistics:-

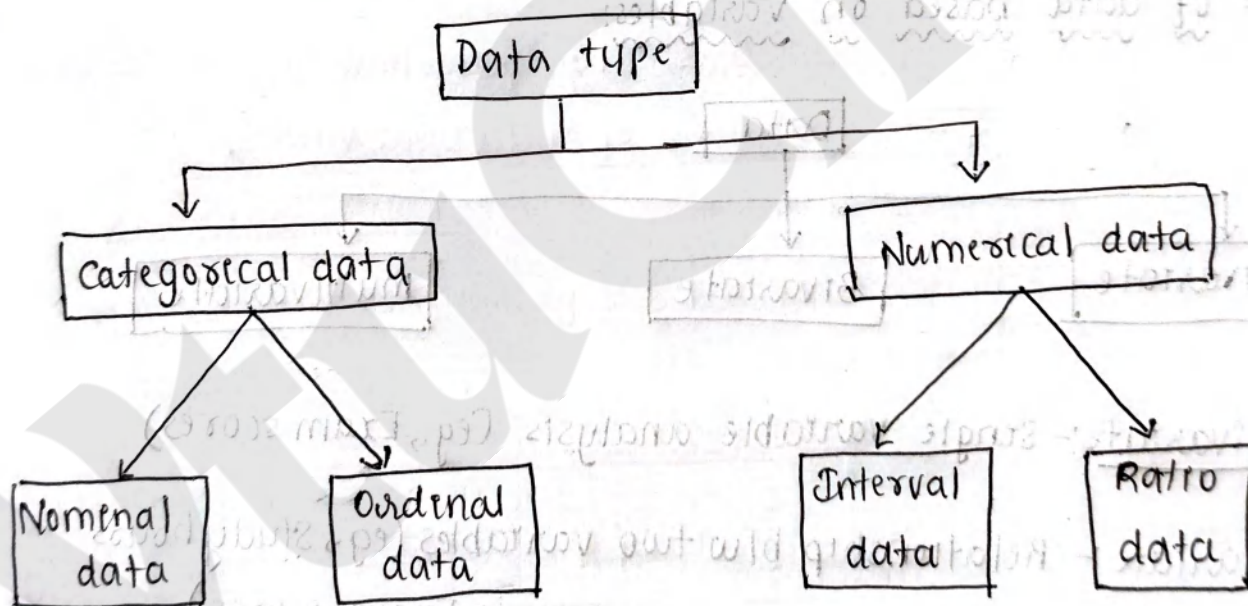
*> It is a branch of statistics that summarizes and describes data without analyzing machine learning projects.

*> Data visualization is a branch of study that is useful for investigating the given data. Uses plots to present and explain data.

*> Exploratory Data analysis (EDA) — combines descriptive statistics and visualization to understand data before applying ML.

Dataset and data types:-

A dataset consists of multiple data objects, each containing attributes. Attributes define characteristics like name, age (or) test results.



1) Categorical data:-

*> Categorical data represents labels or categories. It does not involve numerical values.

i) nominal data:- No specific order (eg., Gender, Blood type)

ii) ordinal data:- Has a meaningful order but difference b/w values are not measurable (eg. Education level : High school, Bachelor's, Master's)

2) Numerical data:-

Numerical data represents measurable values.

i) Interval data:- Has meaningful differences but lacks a true zero.
(eg., Temperature in celcius)

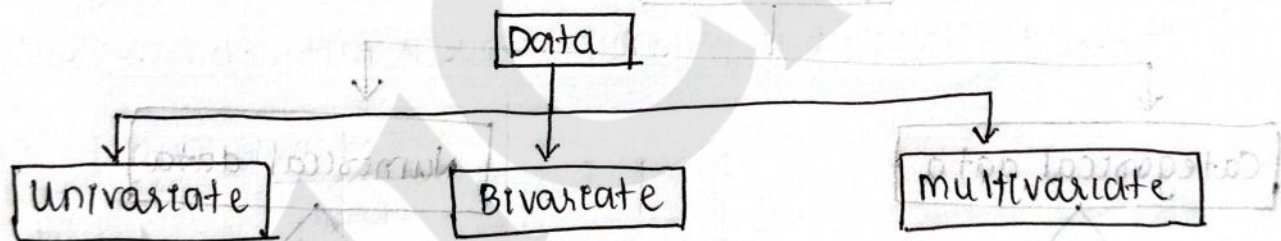
ii) Ratio data:- Has a true zero and meaningful differences.
(eg., weight, height)

Other classifications:-

i) Discrete data:- whole numbers, countable values (eg. no of students in class)

ii) Continuous data:- Can take any value within a range, including decimals (eg. Height, time)

Types of data based on variables:-



*> Univariate:- Single variable analysis (eg., Exam scores)

*> Bivariate:- Relationship b/w two variables (eg., study hours v/s Exam scores)

*> Multivariate:- More than two variables (eg., Exam scores, study hours & sleep duration).

Univariate Data Analysis and visualization:-

*> Univariate analysis examines one variable at a time. It helps describe data, find patterns and understand frequency distribution, central tendency and variation.

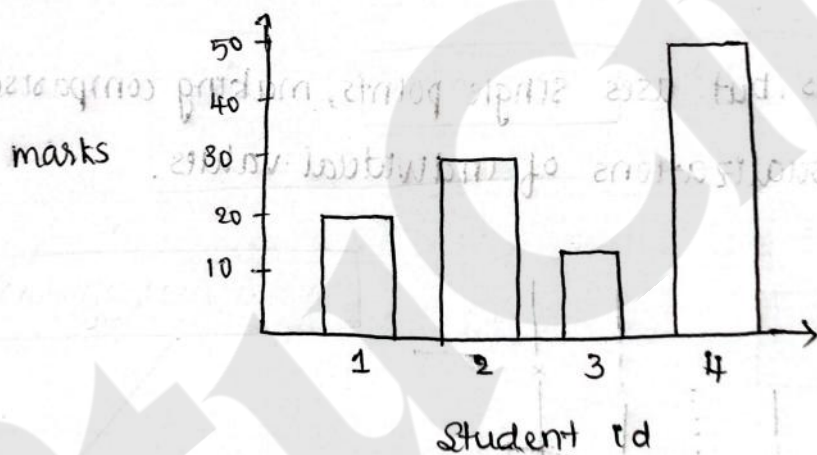
*> Data visualization makes data easier to understand and compare using different types of graphs.

Types of graphs:-

1) Bar chart:-

*> Represents frequency distribution for discrete data, useful for comparisons.

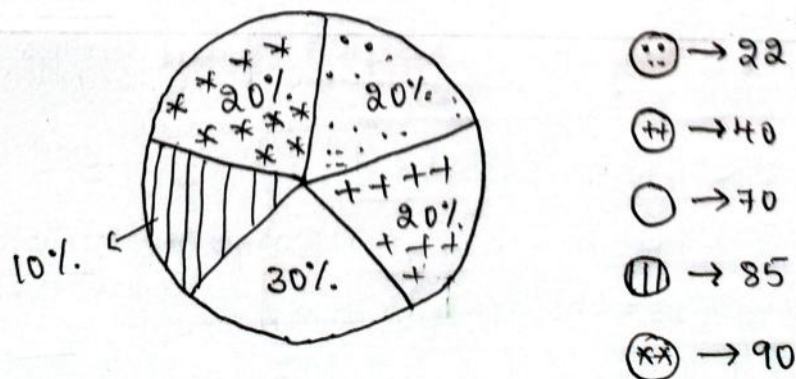
Ex:- Student marks.



2) Pie chart:-

*> Displays percentage distribution of categories in a circular format.

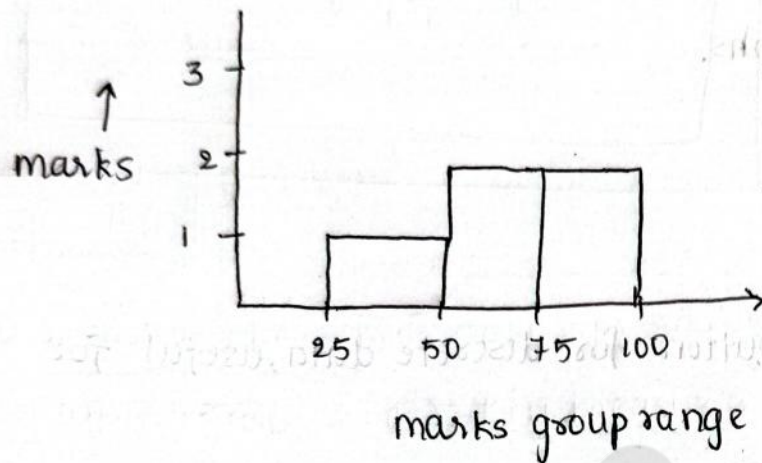
Ex:- Marks distribution in a class. {22, 22, 40, 40, 70, 70, 70, 85, 90, 90}



3) Histogram:-

*> Shows the frequency of numerical data in intervals, highlighting trends and peaks.

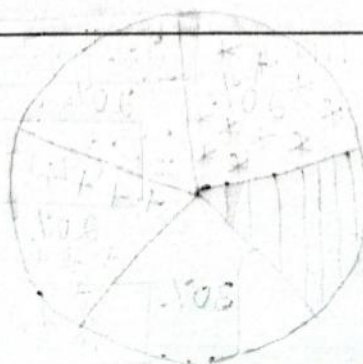
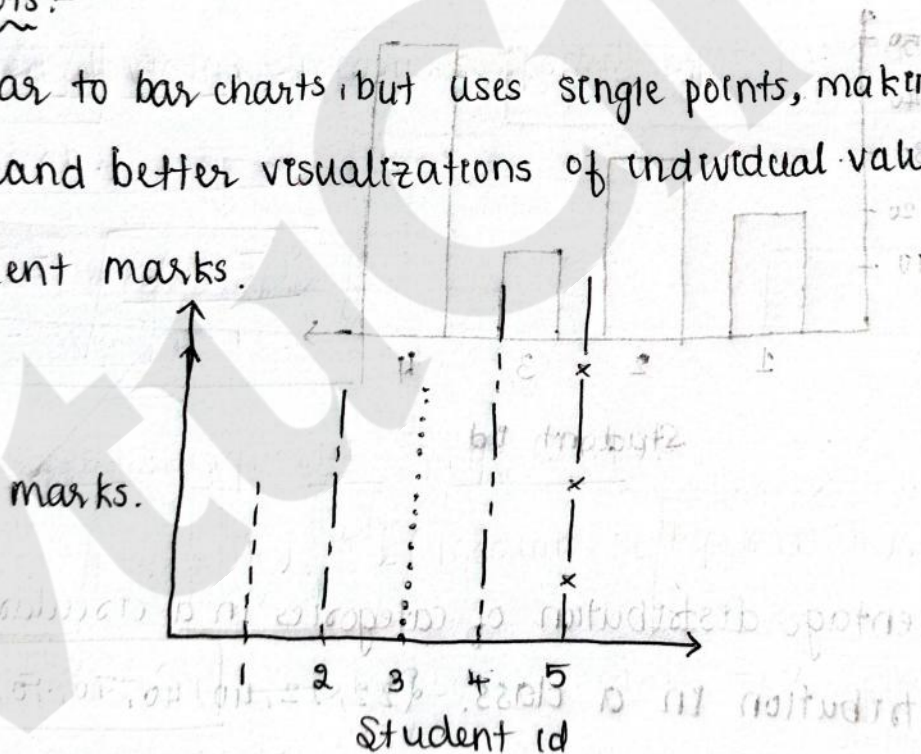
Ex: marks group range.



4) Dot plots:-

*> Similar to bar charts, but uses single points, making comparisons easier and better visualizations of individual values.

Ex: Student marks.



Central tendency:-

* Central tendency summarizes data by identifying a central value.

The 3 main measures are Mean, Median and Mode.

i) Mean (Average):-

* Arithmetic mean: Sum of all values divided by the total count.

$$\Rightarrow \bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n} = \sum_{i=1}^n x_i$$

* Weighted mean: Gives different weights to values based on importance.

* Geometric mean: n^{th} root of the product of n values.

$$\Rightarrow GM = \sqrt[n]{x_1 \cdot x_2 \cdot \dots \cdot x_n} = \left(\prod_{i=1}^n x_i \right)^{1/n}$$

ii) Median:-

* The middle value when data is ordered

* If odd count $\rightarrow$ middle value. If even count $\rightarrow$ avg of 2 middle values

* Formula for continuous data.

$$\text{Median} = L + \left(\frac{\frac{N}{2} - cf}{f} \right) \times \frac{i}{f}$$

Where, L = lower limit of median class

f = frequency

cf = cumulative frequency

iii) Mode:-

* The value that appears most frequently.

* Used for discrete data, as continuous data rarely has repeating values.

* Data can be unimodal (1 mode), bi modal (2 modes) or trimodal (3 modes).

Note:- For large cases,

$$G.M = \frac{\text{anti log } \sum_{i=1}^n \log(x_i)}{n}$$

$$\text{variance} = \sigma^2$$

Dispersion:-

* Dispersion shows how data spreads around the central value

(mean, median, mode). Key measures are:

→ Range:- Difference b/w maximum and minimum values.

→ Standard deviation:- Measures spread by calculating avg distance from mean.

$$\sigma = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}$$

→ Quartiles & Interquartile range (IQR):-

* Q_1 (25th percentile), Q_2 (Median, 50th percentile), Q_3 (75th percentile).

* $IQR = Q_3 - Q_1$ → measures middle 50% spread.

* Outliers → values beyond $1.5 * IQR$ from Q_1 or Q_3

that is., $\text{value} < Q_1 - 1.5(IQR)$

and

$\text{value} > Q_3 + 1.5(IQR)$

are considered as outliers.

Note: Semi-Inter quartile range $(SIQR) = \frac{IQR}{2}$

Five-point summary & box plots:-

$\langle \text{minimum}, Q_1, \text{Median}, Q_3, \text{Maximum} \rangle$

*> Box plot visualizes data spread:

→ Box → Middle 50% (Q_1 to Q_3)

Line inside box → median

whiskers → data range (excluding outliers)

If the median is not centered, data is skewed.

Q) Given, $S = \{5, 10, 15, 20, 25, 30\}$ of marks. Find mean, median, mode, standard deviation and variance, AM, GM.

Sol: mean (μ) = $\frac{5+10+15+20+25+30}{6} = 17.5$

median = $\frac{15+20}{2} = 17.5$

mode = no mode (no repeated value)

$$\sigma^2 = \frac{(5-17.5)^2 + (10-17.5)^2 + (15-17.5)^2 + (20-17.5)^2 + (25-17.5)^2 + (30-17.5)^2}{6-1}$$

= 9.354

var = $(9.354)^2 = 87.497$

AM = 17.5 (AM = mean).

GM = $(5 \times 10 \times 15 \times 20 \times 25 \times 30)^{\frac{1}{6}} = 14.969$

Q) For patient's age test $\{12, 14, 19, 22, 24, 26, 28, 31, 34\}$, find the IQR, and SIQR

Sol: Given, $\{12, 14, 19, 22, 24, 26, 28, 31, 34\}$

$$\text{median} = Q_2 = 24$$

divide into 2 quartiles.

$$Q_1 = \{12, 14, 19, 22\}$$

$$Q_3 = \{26, 28, 31, 34\}$$

↓ median

↓

$$Q_1 = \frac{14+19}{2} = 16.5$$

$$Q_3 = \frac{28+31}{2} = 29.5$$

$$\text{IQR} = Q_3 - Q_1$$

$$\text{IQR} = 29.5 - 16.5$$

$$\text{IQR} = 13$$

$$\text{SIQR} = \frac{13}{2}$$

$$\text{SIQR} = 6.5$$

Q) Find 5-point summary of the test $\{13, 11, 2, 3, 4, 8, 9\}$ and box plot it.

Sol: sort().

$$\Rightarrow \{2, 3, 4, 8, 9, 11, 13\}$$

$$\text{median} = Q_2 = 8$$

$$Q_1 = \{2, 3, 4\}$$

$$Q_3 = \{9, 11, 13\}$$

↓ median

↓

$$Q_1 = 3$$

$$Q_3 = 11$$

$$IQR = 11 - 3$$

$$IQR = 8$$

checking for outliers,

consider Q_1 ,

$$\text{values} \leq 3 - 1.5(8)$$

$$\leq -9$$

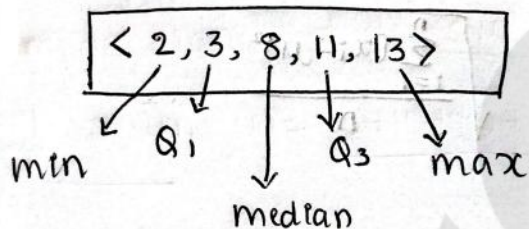
and

$$\text{values} \geq 11 + 1.5(8)$$

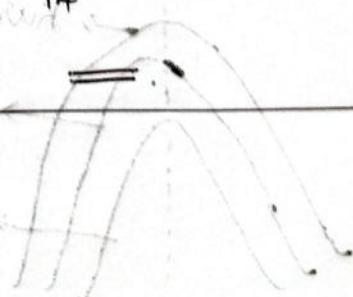
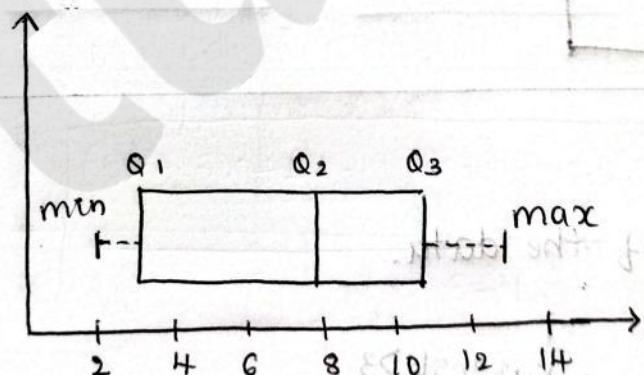
$$\geq 23$$

$\Rightarrow$ no-outliers.

Five point summary:-



Box plot:-



Questions:- (five-point summary & box plot).

⊗ $\{-5, 3, 4, 2, -12, 9, 8, 11, 10, 25, 30\}$

⊗ $\{10, 11, 12, 25, 25, 27, 31, 33, 34, 34, 35, 36, 43, 50, 59\}$

⊗ $\{5, 10, 15, 20, 25, 30\}$

Shape:-

Skewness and kurtosis (called moments) indicate the symmetry/asymmetry and peak location of the dataset.

Skewness (Pearson 2 skewness coefficient):-

$$\frac{3 \times (\mu - \text{median})}{\sigma}$$

where, $\sigma \rightarrow$ population standard deviation

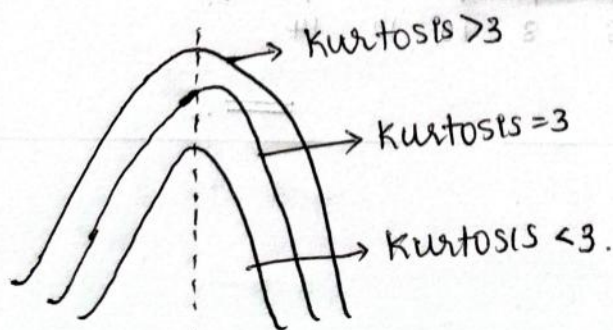
$$\sigma = \sqrt{\frac{\sum_{i=1}^n (x_i - \mu)^2}{n}}$$

General eqn for skewness:-

$$\frac{1}{N} \times \sum_{i=1}^N \frac{(x_i - \mu)^3}{\sigma^3}$$

Kurtosis:-

$\rightarrow$ indicates the peak of the data.



Formula:-

$$\sum_{i=1}^N \frac{(x_i - \bar{x})^4 / N}{\sigma^4}$$

Some other useful measures are:

Mean absolute deviation (MAD):-

*> $\boxed{\text{absolute deviation} = |x - \mu|}$

*> $\boxed{\text{sum of absolute deviations} = \sum |x - \mu|}$

*> $\boxed{\text{Mean absolute deviation} = \frac{\sum |x - \mu|}{N}}$

Coefficient of variation (CV):-

CV is the ratio of standard deviation and mean.

Ex: Given, {90, 80, 90, 90, 70}.

sort().

=> {70, 80, 90, 90, 90}

*> mean = 84

*> median = 90

*> mode = 90

*> AM = 84

*> GM = 83.59

*> $\sigma = 8.944$

*> $\sigma^2 = 80$

*> MAD = 7.2

*> CV = 0.106

*> Skewness = -1.6×10^{-3}

*> kurtosis = 5.07×10^{-4}